

COVID-19 Country-Wise Dashboard & Death Prediction System

Introduction

The COVID-19 pandemic affected countries worldwide and highlighted the importance of data analysis in decision making. This project focuses on analyzing country-wise COVID-19 data and predicting the number of deaths using machine learning. A user-friendly Streamlit dashboard is developed to visualize statistics and make predictions interactively.

Objectives

- To analyze country-wise COVID-19 data
- To visualize confirmed, deaths, recovered, and active cases
- To train a machine learning model for death prediction
- To save and reuse the trained model using Pickle
- To develop and deploy a Streamlit web application

Dataset Description

The dataset used in this project is 'country_wise_latest.csv', which is a public dataset. It contains COVID-19 statistics for multiple countries including confirmed cases, recovered cases, active cases, new cases, new deaths, and total deaths. The target variable selected for prediction is total deaths.

Tools & Libraries Used

Python is used as the core programming language. Pandas is used for data preprocessing and analysis. Matplotlib and Seaborn are used for data visualization. Scikit-learn is used for training the machine learning model. Pickle is used to save and load the trained model. Streamlit is used to build an interactive web dashboard.

Data Preprocessing

The dataset is loaded using Pandas. Relevant features are selected for training the model. The data is divided into training and testing sets using train-test split. This step ensures proper evaluation of the machine learning model.

Data Visualization

A scatter plot between confirmed cases and deaths is created using Seaborn. This visualization helps in understanding the relationship between confirmed cases and total deaths across countries.

Machine Learning Model

Random Forest Regressor is used in this project for prediction. It is an ensemble learning method that provides better accuracy and reduces overfitting. The trained model is saved using Pickle to avoid retraining every time the application runs.

Prediction System

Users can input confirmed cases, recovered cases, active cases, new cases, and new deaths through the Streamlit interface. Based on these inputs, the model predicts the expected number of deaths and displays the result instantly.

Results & Conclusion

The project successfully demonstrates COVID-19 data analysis, visualization, and death prediction using machine learning. The Streamlit dashboard is interactive, easy to use, and fully functional. This project fulfills all the requirements of data preprocessing, visualization, model training, and deployment.

Future Scope

In the future, more machine learning models can be added for comparison. Time-series forecasting and real-time API-based data can also be integrated to improve prediction accuracy.